

Square achievement game on an $n \times n$ grid

OPG-37228

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Problem Two players alternately write O's (first player) and X's (second player) in the unoccupied cells of an $n \times n$ grid. The first player (if any) to occupy four cells at the vertices of a square with horizontal and vertical sides is the winner. What is the outcome of the game given optimal play? Note: Roland Bacher and Shalom Eliahou proved that every 15×15 binary matrix contains four equal entries (all 0's or all 1's) at the vertices of a square with horizontal and vertical sides. So the game must result in a winner (the first player) when $n=15$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>